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Trabajo Práctico N° 7

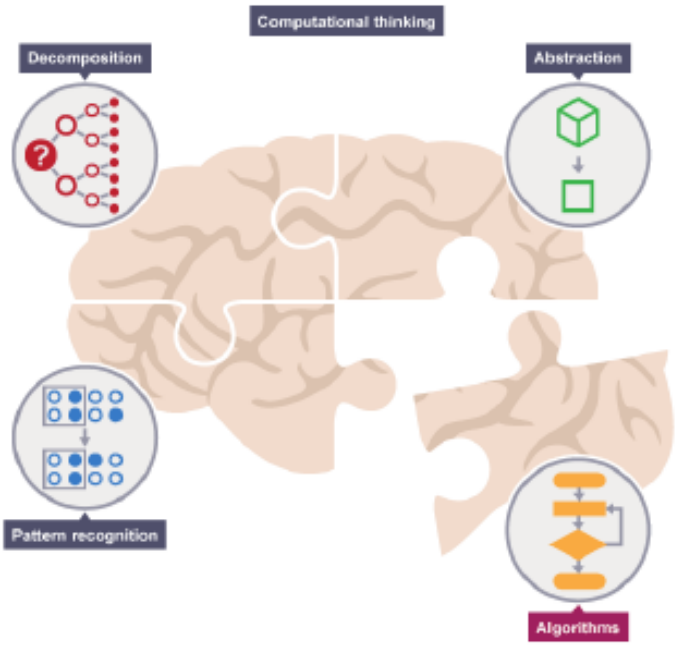
ALGORITHMS

When learning your first programming language, it is easy to get the impression that the hardest part of solving a problem on a computer is translating your ideas into the specific language that will be fed into the computer. This definitely is not the case. The most difficult part of solving a problem on a computer is coming up with the method of solution. After you have developed a method of solution, it is routine to translate your method into the required language, be it Pascal or some other programming language. When solving a problem with a computer, it is therefore helpful to temporarily ignore the computer programming language and to concentrate instead on formulating the steps of the solution and writing them down in plain English, as if the instructions were to be given to a human being. A set of instructions expressed in this way is frequently referred to as an "*algorithm*".

An algorithm is a plan, a set of step-by-step instructions that leads to a solution. Some approximately equivalent words are "recipe", "method", "directions", and "routine". If you can tie shoelaces, make a cup of tea or prepare a meal, then you already know how to follow an algorithm.

In an algorithm, each instruction is identified and the order in which they should be carried out is planned. An algorithm expressed in a language that a computer can understand is called a *program*, which explains why computer languages are called *programming languages*. Instructions may be expressed in a programming language or a human language, like English or French.

The word "algorithm" has a long history, but its meaning has recently taken on a new character]. The word itself derives from the name of the ninth-century Arabic mathematician and astronomer Al-Khwarizmi, who wrote an early and famous textbook on the manipulation of numbers and equations entitled Kitab al-jabr w'almuqabala. The similar-sounding word "algebra" was derived from the Arabic word "al-jabr", which appears in the title of the text and is often translated as "reuniting" or "restoring". The entire title can be translated as "Rules for Reuniting and Reducing". The meanings of the words "algebra" and "algorithm" used to be much more intimately related than they are now. Indeed, until very recently the word "algorithm" usually referred to algebraic rules for solving numeric equations.



Making a plan

It is important to plan out the solution to a problem to make sure that it will be correct. Using computational thinking and decomposition we can break down the problem into smaller parts and then we can plan out how they fit back together in a suitable order to solve the problem. This order can be represented as an algorithm. An algorithm must be clear. It must have a starting point, a finishing point and a set of clear instructions in between.

Ejercicio 1.

Ejercicio 2.

Ejercicio 3.

Ejercicio 4.

Ejercicio 5.